

Gautham Raju

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Education

University of California, Berkeley
Master of Public Policy (MPP)

Berkeley, CA
Aug. 2026 – Present

The University of Texas at Austin
Bachelor of Science in Computer Science; GPA: 3.8/4.0

Austin, TX
Aug. 2019 – May 2023

Experience

Toyota Connected
Software Engineer II

Plano, TX
May 2025 – Aug 2026

- Built a production vehicle-information agent on Amazon Bedrock AgentCore as part of a five-person team, using Python and the Strands Agents SDK to connect an LLM to live data and image services through dynamically discovered Model Context Protocol (MCP) tools and return typed Server-Sent Event streams.
- Engineered the agent runtime with OAuth 2.0 machine-to-machine authentication, short-term conversation memory, retry and timeout handling, and cross-instance Redis/Valkey caching; compressed 100–300 KB tool responses and enabled on-demand detail retrieval to preserve LLM context-window capacity.
- Implemented defense-in-depth AI guardrails through pre- and post-tool hooks that enforce legal-disclaimer retrieval, brand scope, and required calls to action; added periodic compliance reminders and stream-time validation designed to prevent fabricated links and exposure of internal disclaimer codes.
- Developed a configurable evaluation framework for a customer-facing RAG application with scheduled production sanity checks and manual runs across environments; generated synthetic test cases with templates and LLMs with tagged results.
- Implemented asynchronous LLM-as-judge and deterministic metrics for groundedness, query relevance, completeness, contextual precision and recall, coverage, semantic similarity, ground-truth accuracy, retrieval diversity, and disclaimer citation; calibrated thresholds and multi-model aggregation to test legal and organizational guardrails across paraphrases and user personas.
- Operationalized evaluation and agent monitoring through Datadog dashboards, OpenTelemetry tracing, structured conversation logging via AWS SQS, Lambda, and MongoDB, and recurring, programmatically-generated HTML reports for technical and business stakeholders.

Toyota Connected
Software Engineer I

Plano, TX
Jan. 2023 – May 2025

- Automated California Consumer Privacy Act (CCPA) Right to Correct and Right to Delete workflows by designing and implementing a Go service that consumes AWS SQS requests and updates or deletes personal information in MongoDB and Salesforce.
- Researched, implemented, and deployed a STOMP-over-WebSocket service that routes Twilio live-call transcriptions to subscribed Salesforce agent sessions; secured message integrity and authenticity with AWS Key Management Service and reduced average call duration by more than 20 seconds.
- Prototyped a mobile public-safety service with two other engineers, building Java/Spring back-end and Salesforce agent workflows that share real-time location data and support escalation to public safety responders.

Public-Interest Project

MySky

June 2026 – Present

Berkeley Center for Human-Compatible Artificial Intelligence (CHAI); PI: Jonathan Stray

- Built the TypeScript, Lit, and MobX frontend for an open-source ATProto/Bluesky recommender designed to make prosocial feed ranking transparent and user-controllable; created responsive experiences for feed exploration, per-post “Why Am I Seeing This?” explanations, pipeline visualization, and feed-specific settings.
- Designed controls for recommendation-source weights, freshness, and feed purpose, including feed-scoped state, optimistic updates with isolated rollback, live previews of reranked content, pagination, and undo/redo behavior so users can inspect the effects of algorithmic choices before saving them.
- Extended the Python/FastAPI backend with Firebase-authenticated transparency and preference APIs, Bluesky post hydration, and Firestore pipeline snapshots; made candidate time windows, source weights, and purpose configurable per feed and persisted concurrent preference changes through transactional updates.
- Implemented the Bluesky OAuth bridge with Firebase Cloud Functions, PostHog analytics and feedback surveys, CI/CD workflows, and automated unit and end-to-end coverage with Vitest and Playwright.

Skills

AI Engineering and Evaluation: Agentic AI, recommender systems, RAG, MCP, LLM-as-judge, synthetic test generation, semantic similarity, multi-model evaluation, guardrail validation

Programming: Python, TypeScript, Java, Go, JavaScript, SQL, C++, C#

Platforms and Frameworks: Amazon Bedrock, AgentCore, Strands Agents SDK, AWS SQS/Lambda/KMS, Git, Datadog, Docker, Kubernetes, MongoDB, TensorFlow, OAuth 2.0

Community Leadership

Bold Idea: Coding club mentor in underserved communities; technology sustainability committee member

ACEing Autism: Volunteer tennis buddy